

Optimal Controller Design for Two Wheel Mobile Robot

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Abstract— In this paper we worked on Two-Wheel Inverted Pendulum (TWIP). It is highly non-linear, unstable and strong coupling system. Two wheel self balancing robot is based on this concept. During simulation, transfer function is displayed on MATLAB-2016 software. This all work is based on the study of various controllers on Two Wheel Self Balancing Robot (TWSBR). The model is designed on the basis of state space the Mathematical Modeling of Controllers ie LQG & MPC, is well designed and show the comparison in between these two controller. The basic aim of designing this paper is to originate an efficient controller to operate robot in real time environment. The designed LQG controller has better performance and the designed MPC has the best results.

Keywords— Two-Wheel Inverted Pendulum (TWIP), Two Wheel Self balancing Robot (TWSBR), Linear Quadratic Gaussian (LQG), Model Predictive Control(MPC).

I. INTRODUCTION

Two Wheel Self Balancing Robot shows unstable characteristics and just because of these characteristics it has a great scope in research [1-4]. Two wheel self balancing robot is nonlinear system [5]. It is based on fundamental principles of inverted pendulum. It has so many applications such as Rockets, Segway, Self balancing robot, earthquake resistant building design etc. From past few years there are so many research works and available based on the controller design for two wheel self balancing robot, it is all possible just because of its non-linear behavior of inverted pendulum [6]. By using Fuzzy logic PID controller we can easily control a system that is already proposed [7] they used dual PID and LQR method.

Two wheel self balancing robot is built by the researchers at industrial electronics laboratory, and it is based on the inverted pendulum system. We also have seen so many studies about stabilization and optimization of two wheel inverted pendulum system, they use state feedback control with pole placement method [8]. For designing this paper, I go through proportional derivative (PD), Proportional integral derivative (PID), and LQR [7,9], Model predictive control (MPC). So many related research works based on inverted pendulum techniques [12,13] are presented here.

In this paper LQG and MPC design for two wheel self balancing robot is presented in section I, Modeling of system is given in section II, Control techniques of LQG and MPC is provided in section III, simulation results and comparison is in section IV, and at last in section V it is concluded.

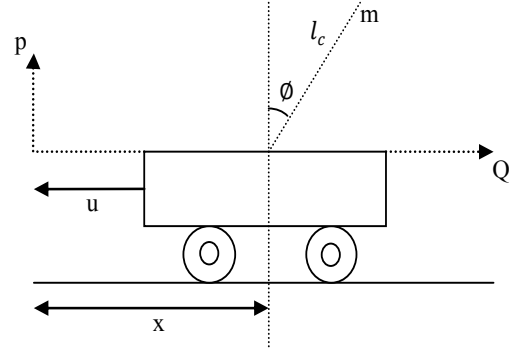


Fig. 1. Two Wheel Mobile Robot

II. MODELLING OF LQG AND MPC

The detailed model of two wheel robot is obtained using inverted pendulum based on cart system. DC motor is used to run and balance this two wheel mobile robot and it plays very important role in two wheel self balancing robot [9-11] that stand on two wheels in Fig-1.

System model is divided into three following part

- DC Motor Model
- Wheel Model
- Model of cart

A. DC Motor Model

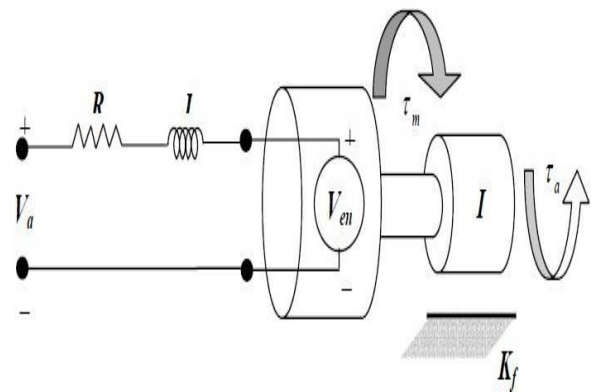


Fig.2. circuit diagram of DC motor

Torque produced by the motor is directly proportional to the current

$$\tau_m = iK_m \quad (1)$$

Voltage is a linear function of velocity of shaft and can be written as

$$V_e = K_e \omega \quad (2)$$

According to Kirchhoff's law

$$L \frac{di}{dt} + Ri = V_a - V_e \quad (3)$$

The equation can be written as

$$\dot{\omega} = -\frac{K_m K_e \dot{\omega}}{I_R R} + \frac{K_m V_a}{I_R R} - \frac{\tau_a}{I_R} \quad (4)$$

In matrix form it can be written as

$$\begin{bmatrix} \dot{\theta} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{K_m K_e}{I_R R} \end{bmatrix} \begin{bmatrix} \theta \\ \omega \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ \frac{K_m}{I_R R} & -\frac{1}{I_R} \end{bmatrix} \begin{bmatrix} V_a \\ \tau_a \end{bmatrix} \quad (5)$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \theta \\ \omega \end{bmatrix} + \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} V_a \\ \tau_a \end{bmatrix} \quad (6)$$

B. Model of wheel of inverted pendulum

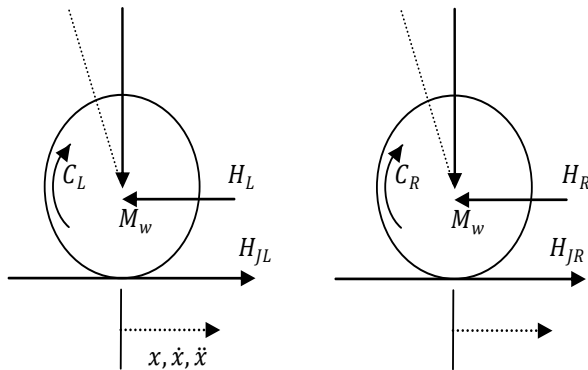


Fig.3. FBD of wheels of robot

In above figure H_L, H_R, P_L, P_R is reaction force that works on different parts of the body, C_L and C_R are the torque to the left and right wheel.

Equation of left wheel and right wheel are

For left wheel

$$M_w \ddot{x} = -\frac{K_m K_e}{Rr} \dot{\theta}_w + \frac{K_m}{Rr} V_a - \frac{l_w \ddot{\omega}}{r} - H_R \quad (7)$$

For right wheel

$$M_w \ddot{x} = -\frac{K_m K_e}{Rr^2} \dot{x} + \frac{K_m}{Rr} V_a - \frac{l_w \ddot{x}}{r^2} - H_L \quad (8)$$

Angular rotation can be formed into linear motion and the equation can be written as

For left wheel

$$M_w \ddot{x} = -\frac{K_m K_e}{Rr^2} \dot{x} + \frac{K_m}{Rr} V_a - \frac{l_w \ddot{x}}{r^2} - H_L \quad (9)$$

For right wheel

$$M_w \ddot{x} = -\frac{K_m K_e}{Rr^2} \dot{x} + \frac{K_m}{Rr} V_a - \frac{l_w \ddot{x}}{r^2} - H_R \quad (10)$$

Resultant equation by adding last two equation will

$$2 \left(M_w + \frac{l_w}{r^2} \right) \ddot{x} = -2 - \frac{K_m K_e}{Rr^2} \dot{x} + 2 \frac{K_m}{Rr} V_a - (H_L + H_R) \quad (11)$$

C. Model of cart

Cart model is the model of inverted pendulum, so that the equation will be

$$(I_p + l^2 M_p) \ddot{\theta}_p - 2 \frac{K_m K_e}{Rr} \dot{x} + 2 \frac{K_m}{Rr} V_a + M_p g l \sin \theta_p = -M_p \ddot{x} l \cos \theta_p \quad (12)$$

$$2 \frac{K_m}{Rr} V_a = \left(2M_w + 2 \frac{l_w}{r^2} + M_p \right) \ddot{x} + 2 \frac{K_m K_e}{Rr^2} \dot{x} + M_p l \ddot{\theta}_p \cos \theta_p \cos \theta_p - M_p l \dot{\theta}_p^2 \sin \theta_p \quad (13)$$

Above equations can be linearized by taking

$$\theta_p = \pi + \varphi \quad (14)$$

Where φ is an angle which is at right angle upside direction.

$$\cos \theta_p = -1, \sin \theta_p = -\varphi, -\varphi \frac{d\theta_p^2}{dt} = 0$$

$$\ddot{x} = 2 \frac{K_m K_e (M_p l_r - I_p M_p l^2)}{Rr^2 a} \dot{x} + \frac{M_p^2 g l^2 \varphi}{\alpha} + 2 \frac{K_m (I_p + M_p l^2 - M_p l_r)}{Rr \alpha} V_a \quad (15)$$

$$\ddot{\varphi} = 2 \frac{K_m K_e (r \beta - M_p l)}{Rr^2 a} \dot{x} + \frac{M_p g l \beta \varphi}{\alpha} + 2 \frac{K_m (M_p l - r \beta)}{Rr \alpha} V_a \quad (16)$$

TABLE I: SPECIFICATION OF THE ROBOT

Parameter	Description	Value	Units
g	Gravitational acceleration	9.83	m/s ²
r	Wheel radius	0.072	M
M _w	Wheel mass	0.5	Kg
M _p	Body mass	5.5	Kg
I _w	Wheel inertia	0.0014	Kgm ²
I _p	Body inertia	0.22	Kgm ²
l	Distance from body's centre of mass	0.2	m
K _m	Motor torque	0.0435	Nm/Amp
K _e	Back EMF	0.0435	V/(rad/s)
R	Terminal Resistance	3.0	Ω

III. CONTROLLER DESIGN

By putting values of different parameters, we can get state space model of plant and the Impulse response of the open loop plant can be shown in fig.4. It can be shown by the figure that it is highly unstable as one of its poles lies in right hand plane.

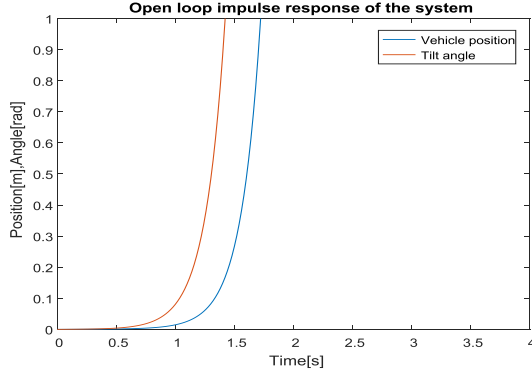


Fig.4. Impulse response of open loop system

When the system is reproduced with the help of controller, then the designed controller tries to upright the inverted pendulum with respect to vertical position angle. The constraint settling time for cart's position as well as pendulum's angle is 4 sec and the range for the pendulum's angle with respect to vertically upright position is ± 0.5 to ± 0.7 radians

A. Linear Quadratic Gaussian Controller (LQG)

Linear quadratic controller is combination of LQR and kalman filter it can be used for both the system linear time variant and linear time invariant. Here we design LQG controller and MPC separately. Plant must be linearized about the origin to obtain the optimal control, linear time system to form a continuous time system will be

$$\dot{x} = Ax + Bu \quad (17)$$

$$y = Cx \quad (18)$$

State feedback law gives the cost function for optimal control approach

$$J = \frac{1}{2} \int_0^\infty (x^T Q x + U^T R U) dt \quad (19)$$

Here:

Q is symmetric positive semi definite matrix

R is symmetric positive definite matrix

Optimal control input is given as

$$U = -Kx \quad (20)$$

Where 'K' is state feedback gain

$$K = R^{-1} B^T P \quad (21)$$

P can be calculated using Riccati Equation

$$A^T P - PA + Q - PBR^{-1}B^T P = 0 \quad (22)$$

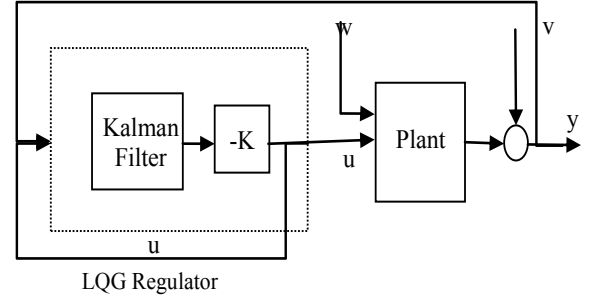


Fig.5. Block diagram of LQG Controller with plant

For regulating the plant state space equation is

$$\dot{x}(t) = Ax(t) + BU + L(y(t) - Cx(t) - DU) \quad (23)$$

$$U = -Kx(t) \quad (24)$$

$$\dot{x}(t) = (A - BK - LC + LDK)x(t) + Ly(t) \quad (25)$$

Here 'L' is Kalman Filter gain

B. Model Predictive Control(MPC)

To design a model predictive control the continuous time Laguerre function and Kautz function have to use. The design model of continuous time predictive control is similar to the discrete-time. Main aim is to design a model predictive control is to model the control trajectory using orthonormal functions.

$$l_i(t) = \sqrt{2p} \frac{e^{pt}}{(i-1)!} \frac{d^{i-1}}{dt^{i-1}} [e^{t-1} e^{-2pt}] \quad (26)$$

p is time scaling factor here that determine their exponential decay rate. By summing the state vector initial conditions

$$L(0) = \sqrt{2p}[1 \ 1 \ \dots \ 1]^T \quad (27)$$

$$L(t) = e^{A_p t} L(0) \quad (28)$$

The augmented state-space model is defined as

$$\begin{bmatrix} \dot{z}(t) \\ \dot{y}(t) \end{bmatrix} = \begin{bmatrix} \overbrace{A_m}^A & \overbrace{o_m^T}^B \\ \underbrace{C_m}_{o_{q \times q}} & \underbrace{o_{q \times m}}^B \end{bmatrix} \begin{bmatrix} z(t) \\ y(t) \end{bmatrix} + \begin{bmatrix} \overbrace{B_m}^B \\ \underbrace{o_{q \times m}}^B \end{bmatrix} u(t) \quad (29)$$

$$y(t) = \underbrace{[o_m \ I_{q \times q}]}_C \begin{bmatrix} z(t) \\ y(t) \end{bmatrix} \quad (30)$$

But for a given τ , matrix $\Phi(\tau)^T$ satisfy following linear algebraic equation

$$A\Phi(\tau)^T - \Phi(\tau)^T A_p^T = -BL(\tau)^T + e^{A\tau} BL(0)^T \quad (31)$$

In the same way to the discrete time case, in continuous predictive control, the cost function that we can assume to take in the form

$$J = \int_0^{T_p} (x(t_i + \tau|t_i)^T Q x(t_i + \tau|t_i) + \dot{u}(\tau)^T R \dot{u}(\tau)) d\tau \quad (32)$$

Here $\dot{u}(\tau)$ is the optimal control that can be easily obtained by minimizing cost function J.

IV. RESULTS AND DISCUSSION

Entire/whole simulation work has been done using MATLAB software. Impulse response obtained with LQG controller is good. Settling for position and pendulum angle is 2.30 s and 1.78 s respectively.

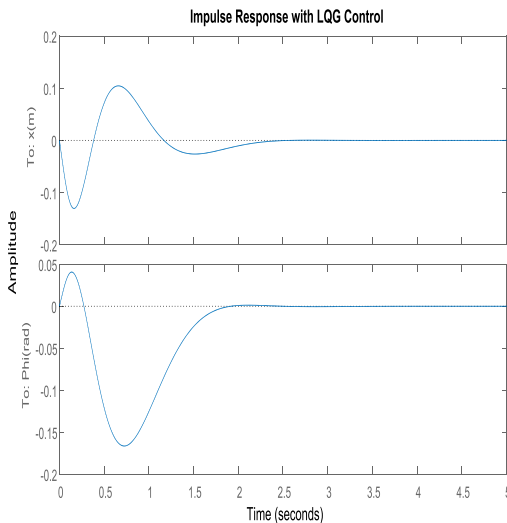


Fig.6. Impulse response of LQG Controller with plant

A train of disturbance is given to robot initially and at after 10 sec, and disturbance rejection capability of LQG controller is found very good.

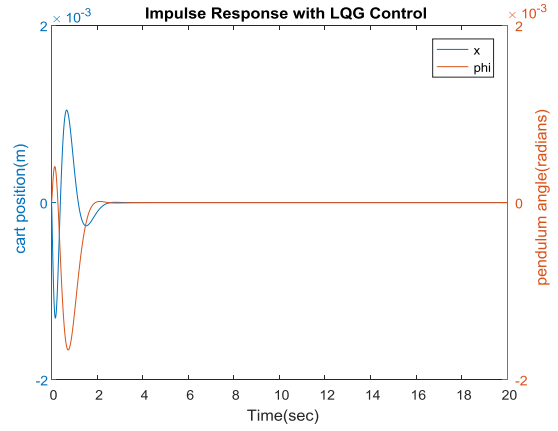


Fig.7. Disturbance rejection capability of LQG Controller with plant

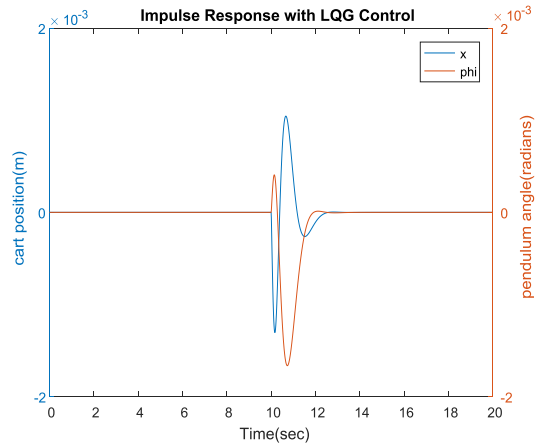


Fig.8. train of disturbance at t=10s & disturbance rejection Capability of LQG Controller with plant

A laguerre function is used to describing the CPMC controller. Various function for different values of p and N are shown in fig. 9.

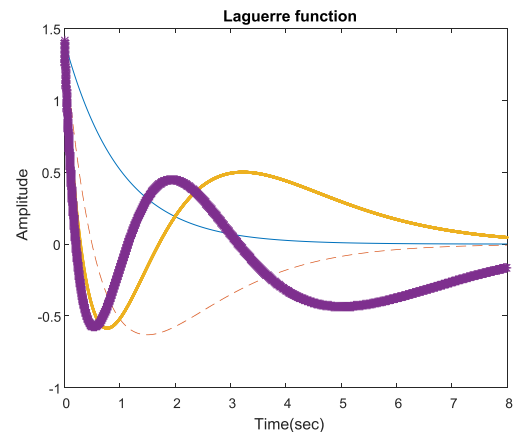


Fig.9. Laguerre function for different values of p & N

The use of laguerre function can be understood by fig. 10 where we use Simpson's rule for calculating co-efficient of laguerre function. It is clearly mentioned that if we increase no. of order for co-efficient then laguerre function completely overlaps the plant and gives satisfactorily good results.

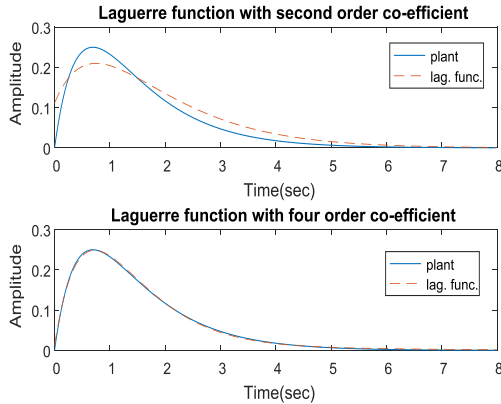


Fig.10. Comparison of Laguerre function with plant for increasing no. of order for co-efficient

Finally we design a CPMC controller and implement it with the help of Laguerre function. Impulse response of MPC controller is shown in fig.11

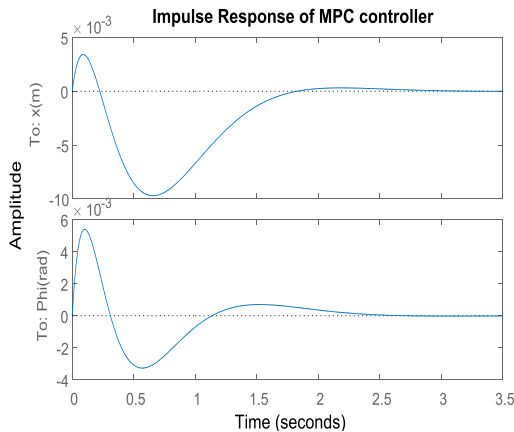


Fig.11. Impulse response of MPC Controller with plant

Settling time for cart's position and pendulum's angle is given in Table 2.

TABLE II COMPARISON OF CONTROLLER

Controller	Settling Time for x (in sec.)	Settling Time for phi (in sec.)	Peak amplitude of phi (in rad.)
LQG	2.3	1.78	0.166
MPC	2.14	1.66	0.0463

V. CONCLUSION

Simulation work have been done using MATLAB program. Impulse response of system using LQG controller and MPC are calculated and both of them compared. Design specification and results of both the controller found well, MPC show much better performance and robustness as compared to LQG.

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